Reg. No. [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [ ] [18MTAPT202]

SECOND SEMESTER M.TECH EXAMINATION, JUNE 2019

DESIGN AND DEVELOPMENT OF AERO ENGINE SYSTEMS AND TESTING

Time: 03 Hours **Maximum Marks:100**

Instructions: Answer five full questions. Question numbers one and two are compulsory

1. a. An axial flow compressor stage has blade root, mean and tip velocities of 150, 200 and 250m/s respectively. The stage is to be designed for a stagnation temperature rise of 20K and an axial velocity of 150 m/s, both constant from root to tip. The work done factor is 0.93. Assuming 50 percent reaction at all radii, calculate the stage air angles at root, mean and tip. What is the disadvantage of this type of design? **08**

   b. The primary-zone airflow pattern by creation of toroidal flow reversal is of prime importance to flame anchoring in aero-gas turbine engine combustion chamber. How is this achieved? Give a brief write-up with illustration by appropriate sketches. **06**

   c. Free vortex principle is commonly applied in the design of a turbine stage. Give a brief account. **06**

2. a. An axial compressor was tested in Albuquerque, New Mexico, on a day when the barometric pressure was 86.4 kPa and the ambient temperature was 27 ^{0}C. When it was running at 15,200 rpm, the measured compressor flow rate was 27.2 kg/s and the delivery pressure was 526.5 kPa. **10**

   i. Determine from these test conditions the shaft speed, mass flow rate and pressure ratio that could be considered equivalent performance if the machine were operated at standard atmospheric conditions (101 kPa and 15 ^{0}C).

   ii. Following a modification, the machine was re-tested in Los Angeles at the same rpm on a day when the ambient air was at 20^{0}C and 102kPa. This time the measured flow rate and the delivery air pressure were 30.1kg/s and 605kPa respectively. Could one consider the modification a success?

   b. Compare commercial part-throttle (Flat-rated) and military full-throttle engines with their typical thrust rating versus ambient temperature curves. **10**

3. a. Explain with diagram the working of fuel pump of single-gear element with centrifugal boost. **10**

   b. Write a note on stake holders in aviation systems. **10**

OR

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4.	a.	Explain with suitable diagrams different kinds of running seals used in gas turbine engines.	10
	b.	Compare the characteristics of vehicle and avionics systems	10
5.	a.	What are the considerations of the engineering processes in the definition phase of product life cycle? Depict them using a block diagram.	07
	b.	Describe the design drivers in the project environment with a block diagram.	07
	c.	Give a depiction of evolution of avionics architectures.	06
		OR	
6.	a.	Write a note on refurbishment phase of product life cycle.	07
	b.	Describe the design drivers in the product environment with a block diagram.	07
	c.	Briefly explain a generic system architecture.	06
7.	a.	Mention the different test methods that are used for providing qualification evidence under verification of system requirements and briefly explain them.	10
	b.	Explain the following:	10
		i. Configuration control and its objectives	
		ii. Forwards compatibility	
		OR	
8.	a.	Write a note on the classical 'V' diagram.	10
	b.	Describe configuration control in Airbus A380 switching network.	10